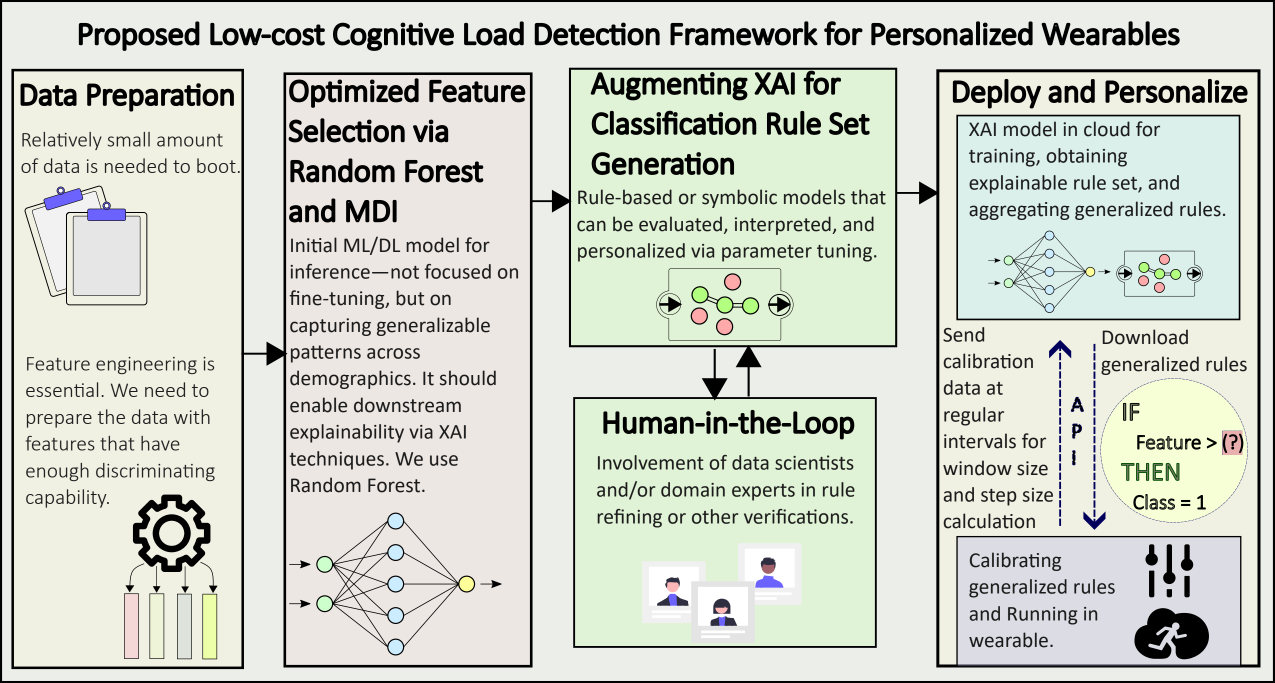
Proposed Low-cost Cognitive Load Detection Framework for Personalized Wearables
Data Preparation
Relatively small amount of data is needed to boot.
Feature engineering is essential. We need to prepare the data with features that have enough discriminating capability.
Optimized Feature Selection via Random Forest and MDI
Initial ML/DL model for inference—not focused on fine-tuning, but on capturing generalizable patterns across demographics. It should enable downstream explainability via XAI techniques. We use Random Forest.
Augmenting XAI for Classification Rule Set Generation
Rule-based or symbolic models that can be evaluated, interpreted, and personalized via parameter tuning.
Human-in-the-Loop
Involvement of data scientists and/or domain experts in rule refining or other verifications.
Deploy and Personalize
XAI model in cloud for training, obtaining explainable rule set, and aggregating generalized rules.
Send calibration data at regular intervals for window size and step size calculation
API
Download generalized rules
IF
Feature > (?)
THEN
Class = 1
Calibrating generalized rules and Running in wearable.